

Integrated workflow – Connecting ImSpector and MACS® iQ View 3D – Large Volume

Lab handout

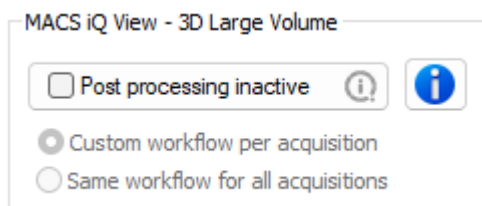
This lab handout explains how to connect the acquisition program ImSpector with the post-processing program MACS® iQ View 3D – Large Volume. Either on the same or on different workstations. This enables an integrated workflow with running a measurements that will automatically be post-processed.

1. Integrated Workflow – Requirements and general information

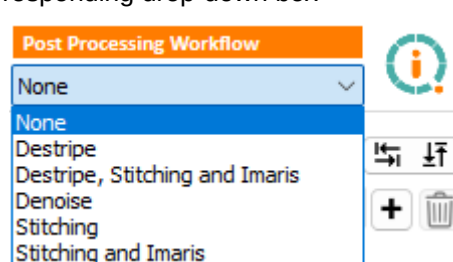
- It is mandatory to have MACS iQ View 3D - Large Volume 1.2 (or higher) running, either on the acquisition PC itself or on a different workstation or server.
- If MACS iQ View 3D is running on a different workstation or server please ensure that both MACS iQ and ImSpector have write access to a shared data directory. Ask your local IT if necessary.
- In case the two programs are not connected, you find a corresponding message at the bottom right of the ImSpector software 7.8 (or higher). See sections 2. and 3. On how to set up connections

 MACS iQ not connected

- If the connection is established, the Experiment Manager includes a new control to activate post- processing:

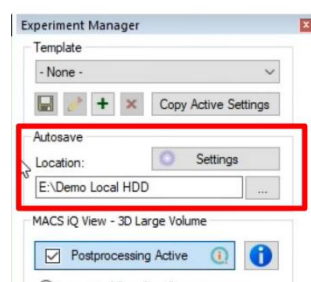


A post-processing workflow can be applied to all measurements or individual workflows can be chosen for each individual measurement listed in the Experiment Manager. A workflow can be selected from the corresponding drop-down box



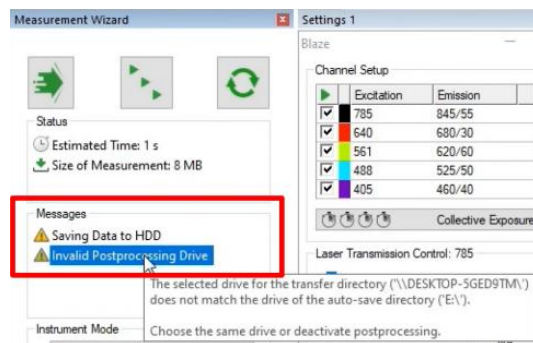
- **Note:** You will only find workflows that are applicable to your current measurement settings! Custom workflows have to be defined in MACS iQ View 3D before they are available in ImSpector.

- Before starting a measurement with the integrated workflow, please make sure that your Autosave directory is within the transfer directory specified in both ImSpector and MACS iQ



View 3D Large Volume.

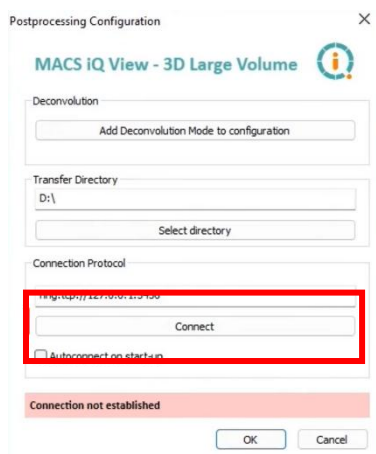
If not, a warning will be displayed in the new Messages window.



- If a workflow was selected for a measurement, the postprocessing job will immediately start once the acquisition has finished. The progress of submitted jobs can be monitored when clicking the info button .

2. Setting up integrated workflows on the acquisition workstation

1. Start both programs.
2. In ImSpector: Navigate to 'Settings - Post-Processing Settings'.
3. Define the transfer directory. Make sure it is an SSD. For checking which drive is an SSD, open the task manager and select "Performance". There you can check the name and type of hard disk.
4. Select connect. You can also enable the 'Auto connect on start-up' option which will automatically establish a connection whenever MACS iQ View 3D - Large Volume is detected.



Note: Once the connection is established, the icon in the lower right corner will change its color and your Experiment Manager tab will display a new control to activate processing.

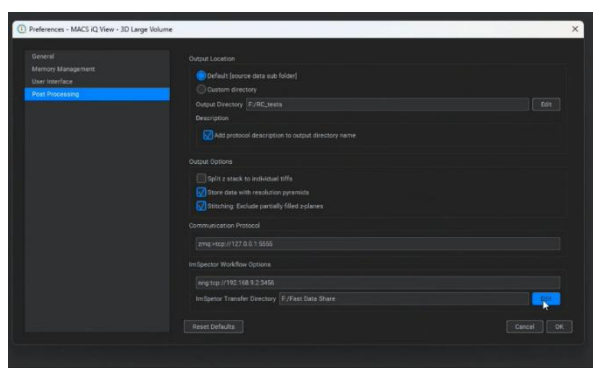


3. Setting up integrated workflows with MACS iQ View 3D - Large Volume on a connected workstation

In case you are using MACS iQ 3D and ImSpector on different workstations, they have to be connected first as described above.

At the graphical workstation

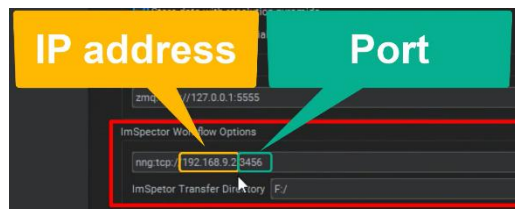
1. Start MACS iQ View 3D on the graphical workstation and navigate to Edit – Preferences – Postprocessing – ImSpector Transfer Directory.



2. Select the shared directory that can be accessed from your acquisition workstation. We recommend to use an SSD RAID for transferring data, especially if you want to transfer your data directly while imaging. For checking which drive is an SSD, open the task manager and select "Performance". There you can check the name and kind of drives.
3. Add the IP address of your graphical workstation:
 - a. Right click on your Windows taskbar to select network and internet settings.
 - b. Select the network adapter in use for the connection (Ethernet), in the case of our graphical workstations, this should always be the 10 Gbit Adapter.
 - c. Copy the IP address to the clipboard and also note it down for later use. The IP address is e.g 192.168.9.2 in the following image:



- d. Close by clicking ok.
4. Go to MACS iQ 3D and insert the IP address in the dialogue, leaving the port as specified before (3456).

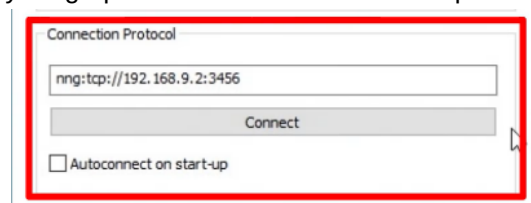


5. Click ok to close the dialogue.
6. Restart MACS iQ 3D. You will be prompted to allow a network connection. Please do so! With this, your graphical workstation is set up.

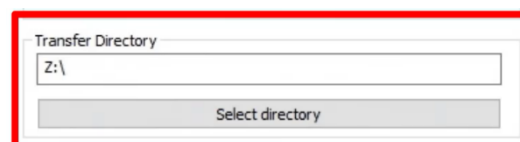
At the acquisition workstation

Next, make sure you can access the shared folder on the graphical workstation from the acquisition workstation.

1. Open ImSpector 7.8.
2. Navigate to Settings – Postprocessing settings.
3. Insert the IP address of your graphical workstation. Leave the port number 3456 unchanged.



4. Select the same transfer directory as chosen in MACS iQ 3D, which on the acquisition PC is a mounted network drive.



5. Select "connect".

Provided the MACS iQ 3D instance is still running on the graphical workstation, the MACS iQ 3D connection icon in the lower right corner of ImSpector 7.2 should now turn green. The connection is established.

